IP address of the domain name you used with the command. For example, this is what the command to look up the IP address of Google.com would look like:

**NSLOOKUP google.com**

The command would generate an output like this:

**Non-authoritative answer:**

**Name: google.com**

**Addresses: 74.125.225.84**

**74.125.225.83**

**74.125.225.81**

**74.125.225.82**

**74.125.225.80**

If NSLOOKUP does not generate a response, you know that either your DNS server is having problems, or your computer cannot communicate with the DNS server due to a connection problem.

## TRACKING NETWORK STATISTICS

The Command Prompt also includes a powerful utility for tracking network statistics – the NETSTAT command. Issuing the NETSTAT command without any switches at the command line displays a list of currently active network connections to your computer. NETSTAT comes with a variety of switches that improve its usefulness:

-/A, which lists all connections and open ports on your computer.

-/E, displays the Ethernet statistics – how many bytes of data your computer has sent and received, and so forth.

-/S, displays the statistics on a per-protocol basis – TCP, IP, ICMP, and others.

NETSTAT is a fairly advanced utility, but you can use it to help track down obscure network problems afflicting your computer.

***

# Chapter 9 – Disk management

Most users don’t think about disk management – they turn on their computers, plug in their flash drives, and expect everything to simply work. However, if you want to install a second internal hard drive in your computer, you'll need to partition and format it. Or you might need to repartition and reformat an external USB hard drive, or (for really advanced users) a USB flash drive.

You can accomplish all these tasks via the Disk Management snap-in in the Microsoft Computer Management graphical application. However, you can just easily do so from the command line. The Command Prompt includes a powerful utility called DISKPART that allows you to manage disks, partitions, and volumes from the command line.

In this chapter we'll show you the basics of using DISKPART.

## TWO CAVEATS

Before we start using DISKPART, there are two points you should keep in mind.

First, DISKPART has to be run as an administrator. You'll need to run DISKPART in a Command Prompt window with administrative rights (we discussed how to do that in Chapter 1). If you try to run DISKPART without administrative rights, you'll only receive the usual "Access is denied" error message.

Second, DISKPART is a powerful utility. With a few keystrokes, you can wipe out entire hard drive partitions - and all the data contained on those partitions. Needless to say, you should take extreme caution while using DISKPART, lest you accidentally destroy large quantities of critical data.

With those points in mind, let's discuss the DISKPART prompt.

## THE DISKPART PROMPT

DISKPART is different from the other commands we've discussed in this book. For most of these commands, you type the command and its options at the prompt, and view the output after the command finishes processing. DISKPART, however, has its own prompt. When you type the DISKPART command, the familiar C:\> prompt will change to the DISKPART prompt, which will look like this:

**DISKPART>**

The normal commands will not work in DISKPART. You will need to use special DISKPART commands, which we will discuss later in this chapter. To leave the DISKPART prompt and return to the normal Command Prompt window, simply type EXIT.

## DISKS, PARTITIONS, VOLUMES

DISKPART works with disks, partitions, and volumes. But what do these terms

mean?

A “disk”, in DISKPART parlance, refers to the actual physical disks themselves. Your hard drive is a disk. So are any USB flash drives, USB hard drives, or optical disks.

A “volume” is a bit more nebulous and harder to define. A “volume”, in Windows, describes a mounted storage device represented by a drive letter. Your C: drive is a volume by this definition. The actual physical disk itself is a “disk” in DISKPART, but the logical filesystem on the disk, along with the drive letter, counts as a volume.

A “partition” in DISKPART is a virtual division on a disk. You could, if you chose, divide a hard drive into four separate virtual parts, and each one of those pieces would count as a partition. Note that a partition is not the same thing as a volume. A volume is formatted with a file system (usually NTFS, but sometimes FAT or FAT32) and has been assigned a drive letter by Windows. A partition is simply a logical division of a physical disk – the partition might be empty, and might not even have been formatted with a file system.

Every volume, however, is placed within a partition. Your hard drive has a partition – most computer hard drives have one big partition that fills the primary hard drive. (Consumer laptop and desktop hard drives generally have one big partition, and a smaller “system restore” partition that contains the Windows reinstallation files in case the computer’s operating system becomes corrupt or virus infected.) To sum up the terms – a physical disk can be divided into partitions, and the partitions can be formatted and mounted as volumes.

## LISTING DISKS, PARTITONS, AND VOLUMES

DISKPART’s commands are context-specific. That means before using a command to alter a disk, a partition, or a volume, you have to first select the volume from the DISKPART prompt. (Needless to say, this makes selecting the proper disk, partition, or volume extremely important!)

But before you can select the appropriate disk, volume, or partition, you need to first employ the LIST command. Using the LIST command, you can find out what disks, volumes, and partitions are on your system. More importantly, you can find out what numbers Windows has assigned those items, since you’ll need to know the numbers in order to select them. So, to find out all the disks on your system, you would use this command from the DISKPART prompt:

**LIST DISK**

The output will look something like this:

**Disk ### Status Size Free Dyn Gpt**

**-------- ------------- ------- ------- --- ---**

*** Disk 0 Online 149 GB 0 B**

**Disk 1 Online 1959 MB 0 B**

In this example, the computer has two disks, Disk 0 and Disk 1. (Computers, using binary, start counting from 0 instead of 1.) Disk 0 is a 149 gigabyte hard drive, while Disk 1 is a two gigabyte (approximately) USB flash drive. Why does Disk 0 have an asterisk character (*) in front of it? That means that Disk 0 is the currently selected disk – any DISKPART commands that affect a disk will affect Disk 0.

To see all the current volumes on your system, use this command:

**LIST VOLUME**

The output will look something like this:

**Volume ### Ltr Label Fs Type Size Status Info**

**---------- --- ----------- ----- ---------- ------- --------- --------**

**Volume 0 D DVD-ROM 0 B No Media**

*** Volume 1 C NTFS Partition 148 GB Healthy System**

**Volume 2 E FAT Removable 1959 MB Healthy**

In this example, the computer has three volumes, Volumes 0, 1, and 2. The output also shows the volume's drive letter in the "Ltr" column, its filesystem in the "Fs" column, the kind of volume (DVD-ROM, Removable disk, etc.) in the "Type", and the size in gigabytes or megabytes in the "Size" column. In this example, Volume 1 is selected – hence the asterisk before its entry in the output. Any commands that affect volumes will affect Volume 1 (in this case, the computer's C drive).

To find the partitions on the currently selected disk, use this command:

**LIST PARTITION**

The LIST PARTITION command will generate an output like this:

**Partition ### Type Size Offset**

**---------- --------------- ------- -------**

*** Partition 1 Primary 1959 MB 31 KB**

Remember, the LIST PARTITION command only lists the partitions on the currently selected disk, not the partitions on every single disk attached to the system. In this example, we are looking at the single partition on the USB flash drive from the earlier examples. As before, Partition 1 (the only partition on the flash drive) is selected, and therefore has the asterisk in front of it.

## SELECTING DISKS, PARTITIONS, AND VOLUMES

Listing disks, partitions, and volumes is well and good, but you'll still need to select them, since using a DISKPART command on the wrong target can prove disastrous. You can select targets of your choosing using the SELECT command. To use the SELECT command, first use the LIST command to view the disks, volumes, or partitions currently available. Make note of the number assigned to each object. For instance, if you wanted to select disk 1, you would use this command:

**SELECT DISK 1**

This will select disk 1. If you use the LIST DISK command afterward, you will see that disk 1 is now the current disk. Selecting volumes works much the same way. If you wanted to SELECT volume 1, use this command:

**SELECT VOLUME 1**

If you run the LIST VOLUME command after this, you will see that Volume 1 is now the selected volume.

Selecting partitions works a little differently. To select a partition, you must first use SELECT DISK to select the disk you want to use, and then LIST PARTITION to see the available partitions on the disk. To then select Partition 0 on the disk, you would use this command:

**SELECT PARTITION 0**

If you then use LIST PARTITION, you will see that Partition 0 is now the selected partition.

Remember that you can only select partitions after you have first selected the disk containing the partition.

## CLEANING DISKS

Once you've selected a disk, you can then issue commands that will affect that disk.

The most drastic command you can use is the CLEAN command. The CLEAN command completely clears off a disk – all volumes and partitions on the disk are deleted. Generally, it's best to use CLEAN when preparing a new disk, or wiping an old one. Make sure you've selected the right disk before using CLEAN – it is quite possible to wipe out valuable data with a single mistyped command!

(Note that the CLEAN command does not overwrite the deleted data – it is still there, and can be removed. To securely wipe a hard drive, you'll need to find a third-party utility that does secure disk erasure.)

## CREATING PARTITIONS

Once you have a blank disk, you can use DISKPART to create partitions on it. You could do that with DISKPART'S CREATE command. You can create five kinds of partitions using CREATE PARTITION, but for this book we will focus on three of them – a primary partition, the extended partition, and the logical partition.

Every disk can have only one primary partition. If a disk has only one partition, it's almost always one big primary partition filling up the drive. Windows's key boot files (like boot.ini and ntldr.exe) must be located on a primary partition in order for the system to boot up.

Likewise, each disk can only have one extended partition. The extended partition is essentially a secondary partition – people will sometimes partition their hard drives so that the system files are located on one partition, and their data files on the second. However, some hard drives have more than two partitions on the disk. How is this accomplished? The answer is logical partitions – each extended partition can contain a larger number of logical partitions. So, if you wanted to divide your hard disk into five separate volumes, you could create a primary partition, an extended partition, and then create the remaining number of logical partitions inside the extended partition.

To create a partition on a disk, first select it with the SELECT command. Then, use the CREATE command to create a partition of the type and size you want. For instance, to create a primary partition:

**CREATE PARTITION PRIMARY**

This command will create a partition that fills up the entire disk. If you want to limit the size, use CREATE with the SIZE modifier. For instance, this command will create a primary partition of 20 gigabytes on the currently selected disk:

**CREATE PARTITION PRIMARY SIZE=20000**

Note that the SIZE modifier uses megabytes for its measurement, and there are (approximately) 1000 megabytes in a single gigabyte. So if you wanted a partition of 20 gigabytes, you would tell the CREATE command to create a partition with a size of 20000 megabytes.

To create an extended partition, use this command:

**CREATE PARTITION EXTENDED**

As with a primary partition, you can also use SIZE to specify the size in megabytes of the desired partition.

To create a logical partition, you'll need to first select an extended partition using the SELECT PARTITION command. Once you've selected the logical partition, use this command to create the logical partition:

**CREATE PARTITION LOGICAL**

As with the previous two examples, you can use the SIZE modifier to limit the space the newly created logical partition will use. This is useful if you want to put more than one logical partition within the extended partition.

## CREATING VOLUMES

After you've create partitions, you can create volumes upon your partitions. DISKPART lets you create several different kinds of volumes, including RAID partitions, but for the purposes of this demonstration we'll stick to creating simple volumes. To create a simple volume, use this command:

**CREATE VOLUME SIMPLE**

You can also use the SIZE modifier to regulate the size of the simple volume. For instance, to create a simple volume with a size of 20 gigabytes:

**CREATE VOLUME SIMPLE SIZE=20000**

As before, the size is measured in megabytes.

## FORMATTING VOLUMES

To "format" a disk means to prepare it with a filesystem for use with your computer. A disk (usually) doesn't come with a filesystem preinstalled - it's simply a blank slate. Formatting the disk with a filesystem marks the disk with a storage system, which means your computer can then read and write data to the disk.

In DISKPART, you format volumes, not disks or partitions. To format a volume,

first select it with the SELECT command, and then use the FORMAT command. The FORMAT command comes with a few options to alter its performance. The most commonly used one is the FS options, which lets you specify which filesystem to use with the volume. For instance, to format the volume with the NTFS filesystem:

**FORMAT FS=NTFS**

You can also select the FAT or FAT32 filesystems. Why would you use a different filesystem than NTFS? Generally, any internal hard disks on your computer should be formatted with NTFS (and any partition where Windows is installed has to be formatted NTFS). Using NTFS on your internal hard disks let you make use of NTFS permissions to secure your files. If you have a USB flash drive, however, it's often best to format it with FAT or FAT32. Mac OS X and Linux computers generally can read from, but not write to, NTFS volumes, and if your flash drive is formatted NTFS Mac and Linux systems will not be able to write data to it. (Of course, for certain secure implementations, you may want to format your USB flash drive with NTFS.)

## ASSIGNING DRIVE LETTERS

Once you've created volumes and formatted them, you can use DISKPART to assign the volumes drive letters. You can do this with the ASSIGN command. To use it, select the volume you that will receive a letter. Once you've selected the appropriate volume, use this command to assign it (for example) the drive letter H:

**ASSIGN LETTER=H**

This will assign the drive letter H to the selected volume.

You can use this command to change the letter of already created volumes - just select the appropriate volume with the SELECT command and then use the ASSIGN command to change the letter. However, take care when doing this, since you might accidentally alter file paths that a program needs to use - Windows Backup, for instance, might become annoyed that the path of your backup hard drive has changed from E:\ to F:\.

Note that you cannot change the drive letter of the volume containing the Windows pagefile. In most cases, the pagefile is located on the same volume as the Windows system files.

## DELETING PARTITIONS

Deleting partitions from DISKPART is easy. It is so easy, in fact, that you should exercise extreme caution while doing so, lest you accidentally destroy important data in the process. You can delete partitions using the DELETE PARTITION command. If you wanted to delete Partition 1 on Disk 0, first select Disk 0, and then Partition 1, using the SELECT command. Once the appropriate partition on the correct disk has been selected, use the DELETE PARTITION command:

**DELETE PARTITION**

This will delete Partition 1 on the currently selected disk. If the partition is an extended partition, any logical partitions within it will be deleted as well.

Note that you must first select the correct disk before deleting any partitions upon it. It's best to be very careful about this - if you accidentally delete Partition 1 on Disk 1 instead of Partition 1 on Disk 0, you run the risk of destroying important data.

## DELETING VOLUMES

Deleting volumes is somewhat less destructive than deleting partitions, and runs less risk of accidentally wiping out large chunks of data (provided, of course, you choose the correct volume). To delete a volume, first select the volume you want to delete with the SELECT command, and then use the DELETE VOLUME command:

**DELETE VOLUME**

This will delete the selected volume.

As always, take care when deleting volumes, and double-check that you have selected the correct volume before deleting it.

***

# Chapter 10 - Connecting to file shares

As we mentioned in Chapter 8, the Windows Command Prompt is based off the old DOS command line, and DOS had barely any networking commands. Starting with Windows for Workgroups, Windows focused heavily on networking, and also on file sharing. "File sharing" is one of the most basic functions of computer networking. It basically means making files available to other users and systems via a network connection. Shared folders are generally referred to as "file shares" or more simply as "shares." Microsoft has for years made it easy to share files from Windows (maybe too easy, given how quickly viruses spread over the Internet in the first years of the 21st century).

But do you need to use Windows's graphical tools to access file shares? Not at all! DOS might have lacked file-sharing commands, but the Windows Command Prompt includes them. Using these commands, you can connect and disconnect from shared folders, and even create your own shares.

## MAPPING NETWORK DRIVES

You might have heard people talk about "mapping a network drive." What does that phrase mean? "Mapping a network drive" refers to taking a shared network folder it and making it appear on your Windows system with a drive letter. When you plug in a USB flash drive or a USB hard drive, you're used to it appearing in the Computer window with the next available drive letter. By mapping a network drive, you can make it appear in Computer with a drive letter, just like any other external drive. You can then issue commands from the Command Prompt to move, delete, and copy files to and from the network share.

Assuming, of course, that you have the proper share permissions, which we shall discuss in the next section.

## SHARE PERMISSIONS

We've already discussed NTFS permissions in Chapter 5. Share permissions, however, are something entirely different from NTFS permissions. NTFS generally deal with which users and groups can access a file. Share permissions control which users can access the shared folder or file, and what level of access they have to the folder, over a network connection. However, NTFS permissions can override share permissions. If you have the share permission to add files to a shared folder, but your account only possesses the NTFS Read permission, you will not be able to copy any files to the share.

This can seem confusing, but taken piece by piece, share permissions are straightforward. In fact, there are six major NTFS permissions, but only three share permissions: Read, Change, and Full.

The Read share permission gives a user permission to view the file or folder over the network. It works a lot like the Read NTFS permission. A user with the Read permission will be able to see the file in the directory, open up the file, and read it. However, the user will not be able to alter the file or delete it.

The Change share permission gives the user permission to alter a shared file. Basically, it functions like the NTFS Write permission. The user has permission to change a file with the Write permission, but he cannot delete the file, and he cannot change its share permissions.

The Full share permission, like the NTFS Full Control permission, gives the user complete control over the file or folder. If you have the Full share permission, you can read the file, make changes to it, and delete it. You can also alter its share permissions for other users.

What happens when the share permissions and NTFS permissions conflict? The rule of thumb is that the most restrictive permission wins. For instance, say you have the Full share permission to a file, but only the NTFS Read permission. You'll be able to read the file, but the NTFS permissions will block you from editing the file in anyway. Likewise, let's say you have the NTFS Full Control permission for a file, but only the Read share permission. You'll be able to read the file, but since your share permissions are limited to Read, you won't be able to do anything else with it.

## VIEWING FILE SHARES

The first step to connecting to file shares from the Command Prompt is to find those shared folders. To connect to a file server, you'll need to know either its IP address, or its Fully Qualified Domain Name. (A FQDN is a computer's proper DNS name - for instance, fileserver.testdomain.com.) Whether connecting via the IP address or the FQDN, you'll need to use a syntax called UNC - Uniform Naming Convention. UNC is a technology Microsoft uses in Windows to designate the location of a network resource. For example, if you have a file server at 192.168.1.1 which contains a shared folder named Data, the UNC path to the server and the folder would look like this:

**\\192.168.1.1\data**

You'll need to use UNC to designate the server and folder you want to use when working with file shares from the Command Prompt. Let's say you want to view

the shared folders available on a server with the IP address of 192.168.1.1. To do so, you would use the NET command. The NET command is the Swiss army knife of the Command Prompt. It does a variety of different things, and is your chief tool for connecting to file shares from the command line. More specifically, you would use the NET VIEW command to view the available file shares on a server:

**NET VIEW \\192.168.1.1**

The command will generate an output that should look something like this:

**Shared resources at \\192.168.1.1**

**Share name Type Used as Comment**

**----------**

**images Disk**

**data Disk**

**backup Disk**